

LINMA2222 - Stochastic Optimal Control & RL

Optimal Portfolio Strategy

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UCLouvain

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1. Modeling and Analysis (Part I)

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Average-reward trading control:

$$\max_{\pi} \lim_{T \rightarrow \infty} \frac{1}{T} * \mathbb{E} \left[\sum_{t=0}^{T-1} r_t \right]$$

State: $x_t = (q_t, z_t^a, z_t^u)^\top$ **Control:** u_t **Noise:** $\xi_t^a, \xi_t^p \sim \mathcal{N}(0, 1)$ **Constraint:** $q_t \in [0, 1]$ **True dynamics**

$$q_{t+1} = q_t + u_t$$

$$z_{t+1}^a = (1 - \omega^a) z_t^a + \omega^a \sigma^a \xi_t^a$$

$$z_{t+1}^u = (1 - \omega^u) z_t^u + \omega^u \beta^u u_t$$

Price / execution and P&L

$$p_{t+1} = p_t + z_t^a + z_t^u + \gamma^u u_t + \sigma^p \xi_t^p$$

$$\bar{p}_{t,t+1} = \theta p_t + (1 - \theta) p_{t+1} + \theta \gamma^u u_t$$

Reward: Profit / Loss g_t

$$g_t := 1000 \cdot \left(\underbrace{(q_{t+1} p_{t+1} - q_t p_t)}_{\text{portfolio value change}} - \underbrace{(q_{t+1} - q_t) \bar{p}_{t,t+1}}_{\text{cash flow}} \right)$$

$$r_t := c(g_t) = \max \left(g_t - \frac{1}{2} g_t^2, 1 - \exp(-g_t) \right)$$

1. Modeling and Analysis (Part I)

— 1.1 Dynamics & reward

- g_t : Net profit/loss from trading
- **Change in portfolio value at market price**
- **Cash buy / sell**

1.2 Baseline linear policy π_{cl} and π_{pcl}

baseline policy

$$\pi_{\text{cl}}(x_t) = K_{\text{cl}}x_t = -0.5q_t + 0.5z_t^a + 0.5z_t^u$$

$$\pi_{\text{pcl}}(x_t) = \max\left(-q_t, \min\left(1 - q_t, \pi_{\text{cl}}(x_t)\right)\right)$$

Average rewards:

- π_{cl} : **0.011451**
- π_{pcl} : **0.005796**

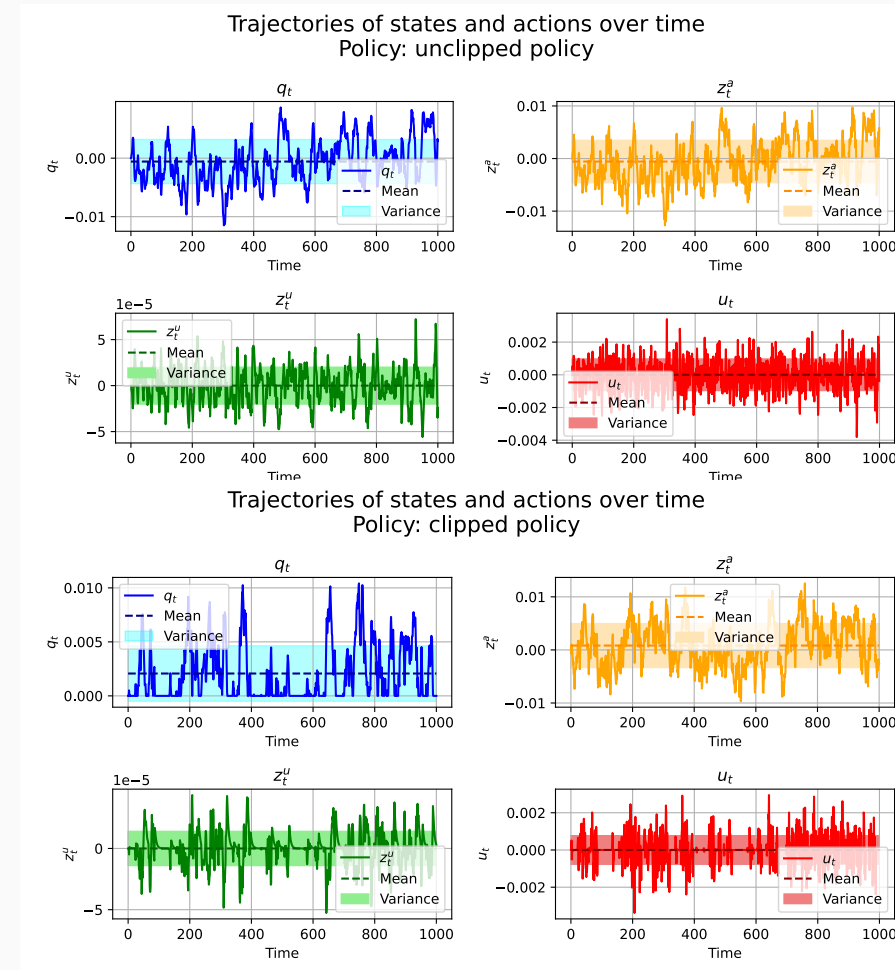


Figure 2: π_{cl} and π_{pcl} : Example trajectory

1. Modeling and Analysis (Part I)

— 1.2 Baseline linear policy π_{cl} and π_{pcl}

- can make $q_t < 0$
=> infeasible

CMA-ES linear policy

$$\pi_{l(x)} = \text{clip}(p_3^\top [q, z^a, z^u], [-q, 1 - q])$$

CMA-ES quadratic policy

$$\pi_{q(x)} = \text{clip}(p_{10}^\top \varphi(x), [-q, 1 - q])$$

where $\varphi(x) = [1, q, z^a, z^u, q^2, (z^a)^2, (z^u)^2, qz^a, qz^u, z^a z^u]^\top$

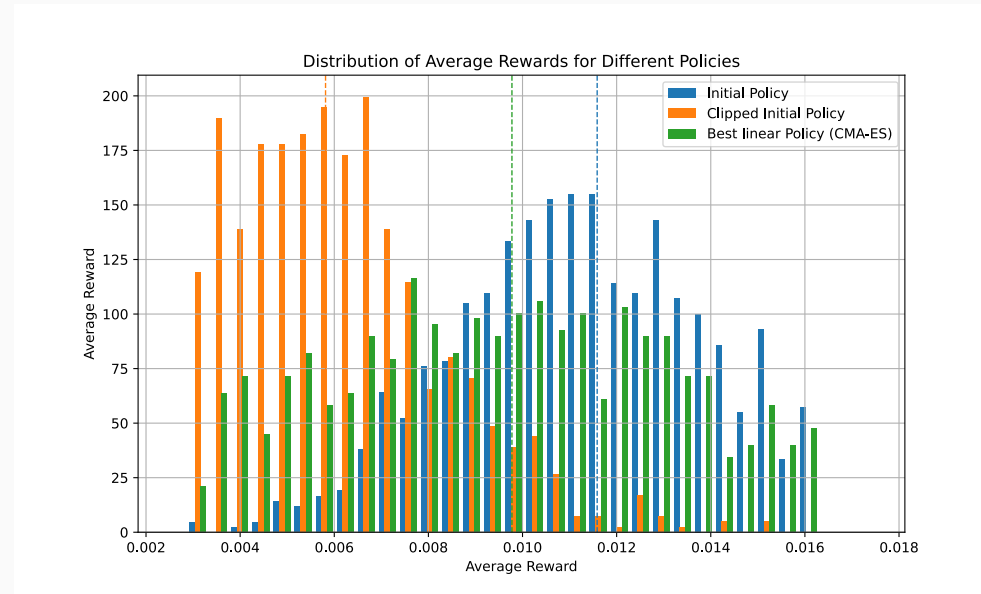


Figure 3: Reward comparison: baseline vs CMA-ES policie

1. Modeling and Analysis (Part I)

— 1.3 Better policy via CMA-ES

Method made for noisy optimization.

- quadratic only slightly better.

2. LQR, LQG and MPC (Part II)

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LQR approximation

Linear dynamics $x_{t+1} = Fx_t + Gu_t + D\xi_t$

Reward: Maximize the expected reward
(minimize cost):

$$\hat{r}(x, u) := \frac{1}{2}x^\top Qx + x^\top Su + \frac{1}{2}u^\top Ru$$

$$\max \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} \hat{r}(x_t, u_t) \right]$$

$$\mathbb{E}[r(x, u)] \approx \mathbb{E}[c_{\text{quad}}(g(x, u))] \approx \mathbb{E}[\hat{r}(x, u)]$$

with $c_{\text{quad}}(g) = g - \frac{1}{2}g^2$

Resulting matrices

$$F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 - \omega^a & 0 \\ 0 & 0 & 1 - \omega^u \end{pmatrix}$$

$$G = \begin{pmatrix} 1 \\ 0 \\ \omega^u \beta^u \end{pmatrix}$$

2. LQR, LQG and MPC (Part II)

— 2.1 LQR model approximation

- linearize dynamics & quadratic reward
- obtain LQR model (F,G,Q,R,S)

LQR policy derivation

Discrete Algebraic Riccati Equation (DARE)

$$M^* = Q + F^\top M^* F - (F^\top M^* G + S)(R + G^\top M^* G)^{-1} (G^\top M^* F + S^\top)$$

Optimal gain

$$K_{\text{lqr}} = -(R + G^\top M^* G)^{-1} (G^\top M^* F + S^\top) \approx \begin{pmatrix} 1.112 & -2.649 & -2.528 \end{pmatrix}$$

Theoretical average reward

$$J^* = -\frac{1}{2} \text{tr}(M^* D D^\top) = 0.01936$$

2. LQR, LQG and MPC (Part II)

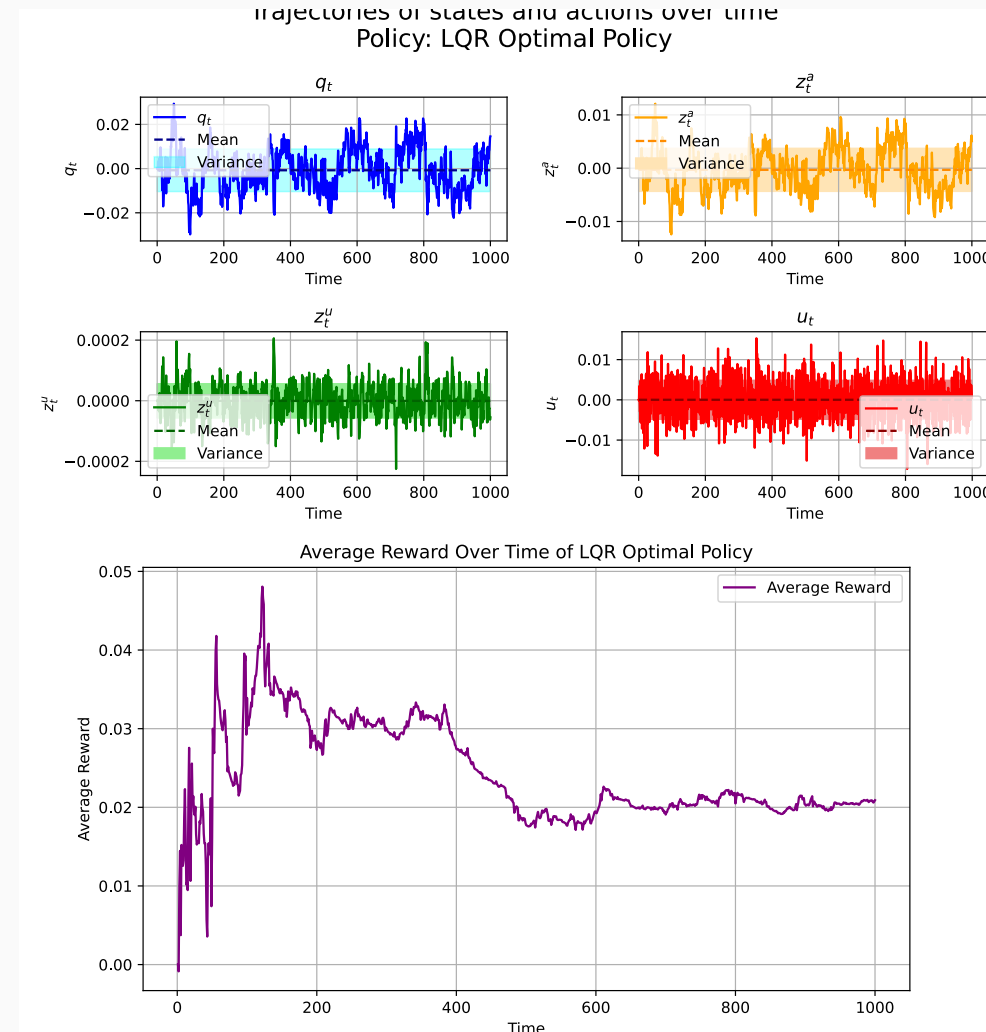
— LQR: Riccati equation & optimal gain

- Solve DARE iteratively or via `scipy.linalg.solve_discrete_are`
- Closed-loop stability: eigenvalues of $(F + GK)$ inside unit circle

LQR policy

$$\pi_{\text{lqr}}(x_t) = -K_{\text{lqr}}x_t$$

- average reward: **0.019467**
 - theoretical : **0.01936**
- average reward: **0.009693** (constrained)



Clipped LQR policy

- average reward decreases

Model Predictive Control

Variables: $z_k \in \mathbb{R}^{n_x}$ (predicted states), $v_k \in \mathbb{R}^{n_u}$ (predicted controls)

Optimization problem at time t :

$$\min_{z,v} \sum_{k=0}^{\mathcal{N}-1} \underbrace{\frac{1}{2} z_k^\top Q z_k + z_k^\top S v_k + \frac{1}{2} v_k^\top R v_k}_{-\hat{r}(z_k, v_k)}$$

Subject to:

$$z_0 = x_t$$

$$z_{k+1} = F z_k + G v_k, \quad k = 0, \dots, \mathcal{N} - 1$$

$$y_{\min} \leq z_{k+1} \leq y_{\max}, \quad k = 0, \dots, \mathcal{N}$$

$$u_{\min} \leq v_k \leq u_{\max}, \quad k = 0, \dots, \mathcal{N} - 1$$

2. LQR, LQG and MPC (Part II)

— 2.3 MPC

- z, q : decision variables (predicted trajectory)
- H, E : output matrix (here H selects position q_t)
- $y_{\min}, y_{\max}$: output bounds (position in $[0,1]$)
- R_0 : terminal cost (here $R_0 = Q$)

Stacked QP formulation

Stack variables:

$$w = [z_0, \dots, z_{\mathcal{N}}, v_0, \dots, v_{\mathcal{N}-1}]^\top$$

Solve :

$$\min_w \quad \frac{1}{2} w^\top H_{\text{obj}} w$$

$$H_{\text{obj}} = \begin{pmatrix} Q & & S & & \\ & \ddots & & \ddots & \\ & & Q & & S \\ S^\top & & R & & \\ & \ddots & & \ddots & \\ & & S^\top & & R \end{pmatrix}$$

Equality constraints :

$$A_{\text{eq}} w = [x_t, 0, \dots, 0]^\top$$

bounds :

$$y_{\min} \leq z_k \leq y_{\max}, \quad u_{\min} \leq v_k \leq u_{\max}$$

2. LQR, LQG and MPC (Part II)

— MPC: QP Reformulation

- Standard QP: efficient solvers available
- Sparse structure exploited
- Receding horizon: only apply v_0^*

- express future states in terms of x_0 and controls V

State prediction: $z_k = F^k x_0 + \sum_{j=0}^{k-1} F^{k-1-j} G v_j$

In matrix form:

$$\underbrace{\begin{pmatrix} z_0 \\ z_1 \\ \vdots \\ z_N \end{pmatrix}}_Z = \underbrace{\begin{pmatrix} I \\ F \\ \vdots \\ F^N \end{pmatrix}}_{\mathcal{F}} x_0 + \underbrace{\begin{pmatrix} 0 & 0 & \dots & 0 \\ G & 0 & \dots & 0 \\ \vdots & \ddots & & \vdots \\ F^{N-1}G & \dots & FG & G \end{pmatrix}}_{\mathcal{G}} \underbrace{\begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{pmatrix}}_V$$

Reduced QP: optimize $V \in \mathbb{R}^{n_u \cdot \mathcal{N}}$

$$\min_V \quad \frac{1}{2} V^\top \hat{H} V + (\hat{F} x_0)^\top V$$

$$\hat{H} = \mathcal{G}^\top \tilde{Q} \mathcal{G} + \tilde{R} + \mathcal{G}^\top \tilde{S} + \tilde{S}^\top \mathcal{G}$$

$$\hat{F} = \mathcal{G}^\top \tilde{Q} \mathcal{F} + \tilde{S}^\top \mathcal{F}$$

where $\tilde{Q}, \tilde{R}, \tilde{S}$ are block-diagonal

Constraints:

input bounds + state bounds via $\mathcal{G}$

2. LQR, LQG and MPC (Part II)

— MPC: Condensed Formulation

- States eliminated $\rightarrow$ QP size reduced from $(n_x + n_u) \cdot N$ to $n_u \cdot N$
- $\hat{H}$ precomputed once; only $\hat{F} x_0$ updated each step

chosen horizon : $\mathcal{N} = 10$

- performance vs computation
- greater N : diverging predictions

Policy	Avg Reward	Feasible	Key characteristic
π_{lqr}	0.019467	✗	optimal for LQR model
Clipped π_{lqr}	0.009693	✓	still good performance
MPC ($\mathcal{N} = 30$)	0.0113082	✓	2x slower than $\mathcal{N} = 10$
MPC ($\mathcal{N} = 10$)	0.0113081	✓	costly
MPC ($\mathcal{N} = 5$)	0.0113076	✓	$\approx$ time $\mathcal{N} = 10$

Takeaways:

- Clipped $\pi_{\text{lqr}} \approx$ MPC performance, but **much faster**

2. LQR, LQG and MPC (Part II)
— Part II Summary: Model-Based Control

- In practice:
 - clipped LQR
 - fast and good performance

3. Policy Improvement (Part III)

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E-PIA on LQR model

Policy iteration steps

1. **Evaluation:** Solve Lyapunov equation for P_k :

$$P_k = Q_k + A_k^\top P_k A_k$$

where $A_k = F + GK_k$, $Q_k = Q + SK_k + (SK_k)^\top + K_k^\top RK_k$

2. **Improvement:** Update gain:

$$K_{k+1} = -(R + G^\top P_k G)^{-1} (S^\top + G^\top P_k F)$$

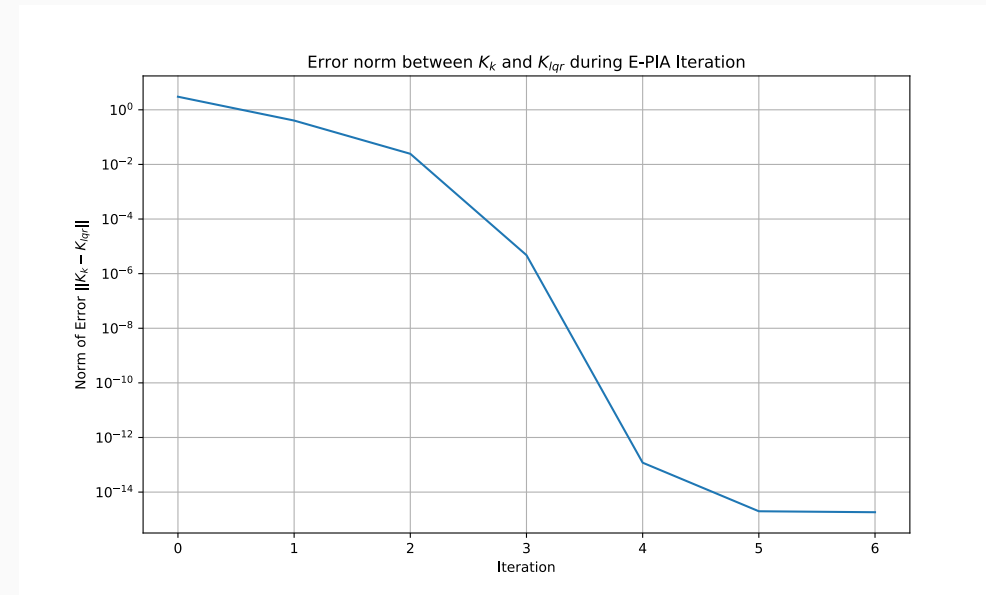
E-PIA convergence

Results

- From π_{cl} , converges to K_{lqr} in ≈ 5 iterations
- Confirms that Riccati-based LQR is optimal for the quadratic model

Learned gain

$$K_{\text{lqr}} \approx (1.112 \quad -2.649 \quad -2.528)$$



LSTD for policy evaluation

Objective Approximate the Q-function under a fixed policy π :

$$Q^\pi(x, u) \approx \theta^\top \psi(x, u)$$

LSTD update Solve for θ using instrumental variables:

$$\theta = \left[\frac{1}{N} W + R_N \right]^{-1} \phi_N$$

where:

$$R_N = \frac{1}{N} \sum_{k=1}^N \Upsilon_k \Upsilon_k^\top \quad \phi_N = \frac{1}{N} \sum_{k=1}^N \Upsilon_{k+1} \gamma_k$$

3. Policy Improvement (Part III)

— 3.2 LSTD: Least-Squares Temporal Difference (Q6.3–Q6.4)

1. $\Upsilon_k = \psi(x(k), u(k)) - \psi(x(k+1), \pi(x(k+1)))$
2. $\gamma_k = c(x(k), u(k))$

LSTD for true model

Key design choices

- **Degree-4 polynomial basis:** cost contains exponential term
- **No bias term**
- **Regularization:** $(R_N + \varepsilon I)$ for numerical stability
- **Large N :** $N = 3000$ samples for well-conditioned matrix

Results

- Good match with Monte-Carlo estimates on true system

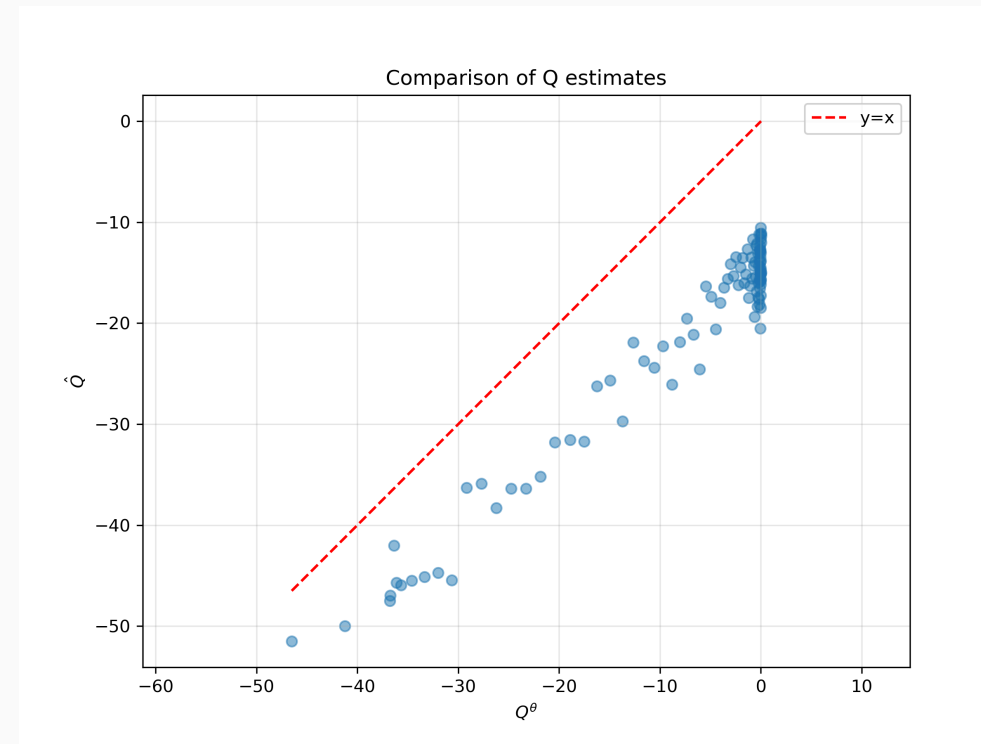


Figure 7: True system: Q^θ vs $\hat{Q}_{MC}$

LSTD for LQR model

Key design choices

- **Degree-2 polynomial basis:** Q_{exact} is quadratic

Results

- Perfect match with the exact Q -function

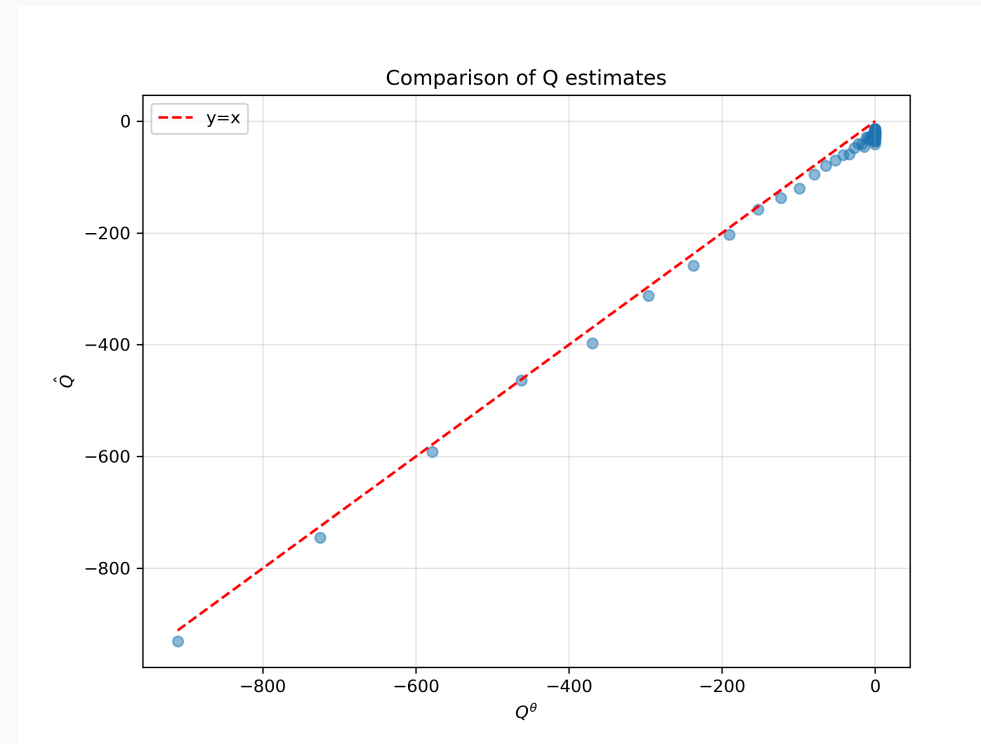


Figure 8: LQR model: Q^θ vs Q_{exact}

LSPI algorithm

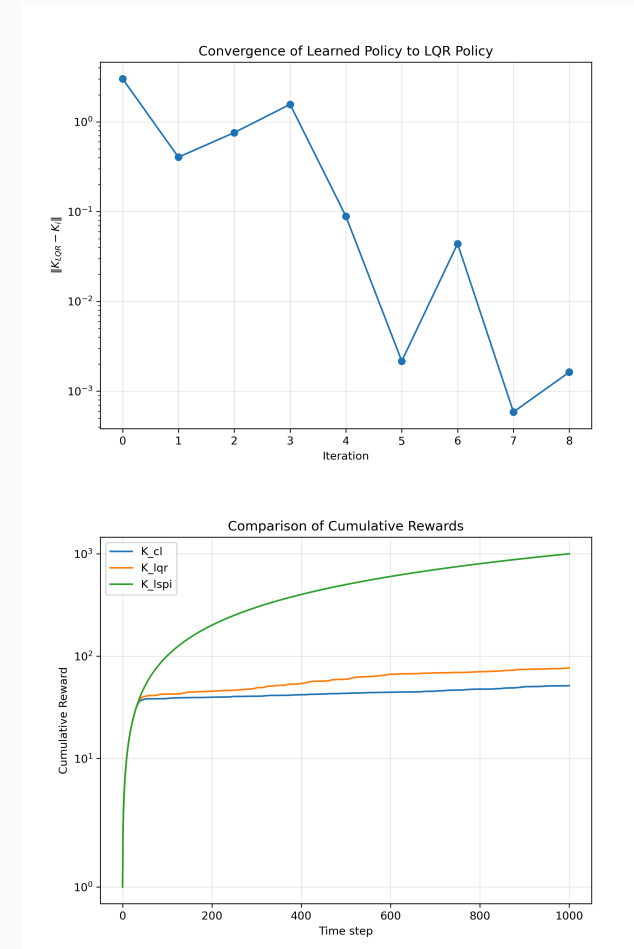
Iterate:

1. **Evaluation:** LSTD $\rightarrow \theta_k$
2. **Improvement:** Find greedy policy w.r.t. Q^{θ_k}

Policy improvement step For linear policy

$$u = -Kx:$$

- Sample states $x_s \in [-1, 1]^3$
- Optimize: $u_s^* = \arg \max_u \theta^\top \psi(x_s, u)$
- Fit: $K_{k+1} = (X^\top X)^{-1} X^\top (-U^*)$



3. Policy Improvement (Part III)

— 3.3 LSPI: Least-Squares Policy Iteration (Q6.5–Q6.7)

Q- λ : online temporal-difference

Algorithm

- Quadratic features: $\psi(x, u) = \text{vec}([x; u][x; u]^\top)$
- $\zeta_{t+1} = \lambda \zeta_t + \psi_t$
- TD error: $\mathcal{D}_t = c_t + Q_{t+1} - Q_t$
- Update: $\theta \leftarrow \theta + \alpha \mathcal{D}_t \zeta_t$

Effect of λ

- $\lambda = 0$: one-step TD (slow but stable)
- $\lambda \rightarrow 1$: Monte-Carlo-like (faster but can diverge)

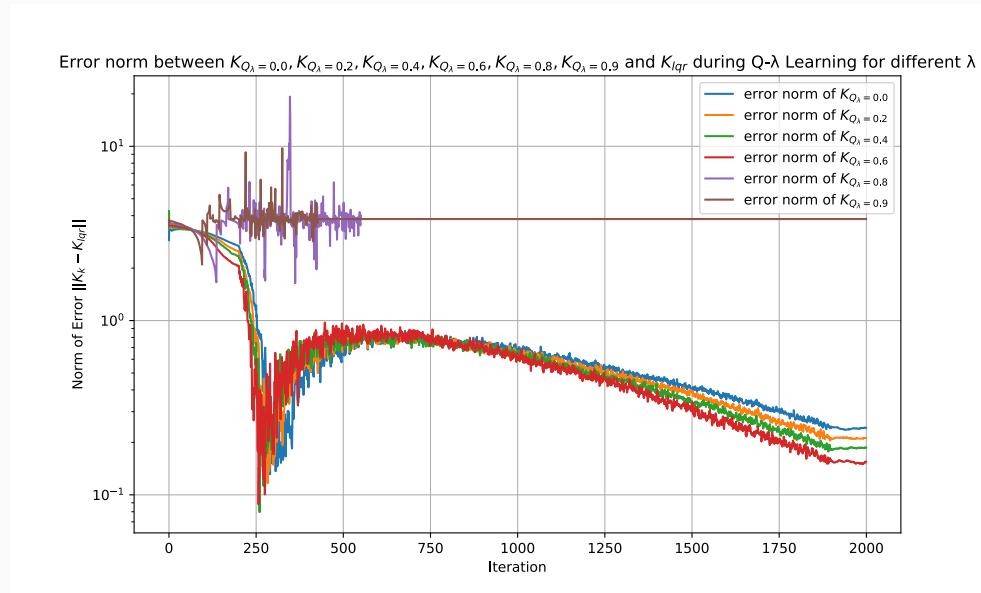


Figure 10: Convergence to K_{lqr} for different λ

3. Policy Improvement (Part III)

— 3.4 Q- λ Learning (Q7)

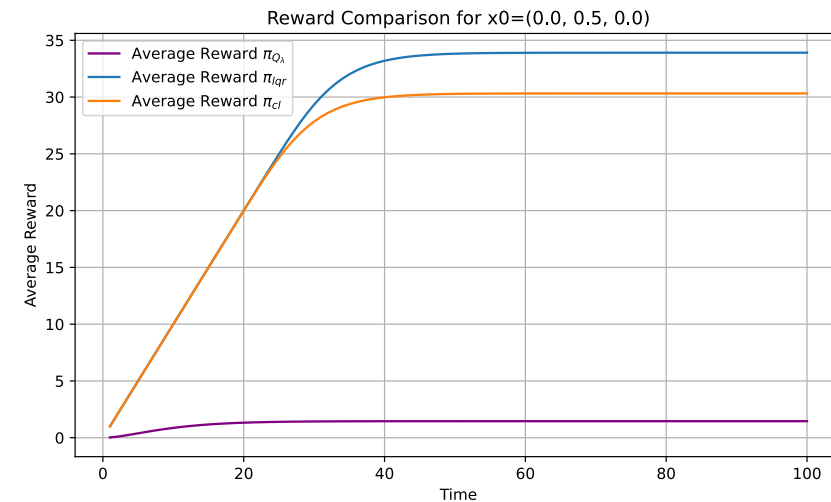
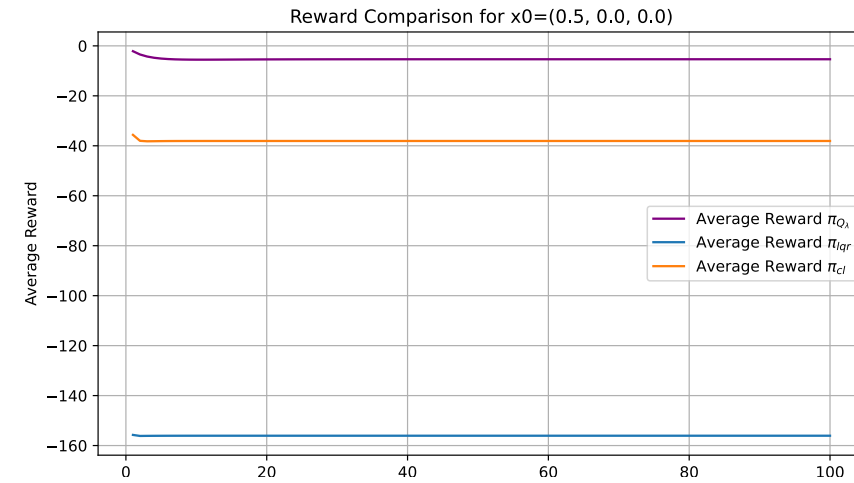
Q- λ performance

On LQR Model

- Converges to K_{lqr} for LQR settings
- Higher $\lambda \rightarrow$ faster convergence (with tuned α)

On deterministic system

- $Q(\lambda)$ converges to $[-0.086, 2.1 \times 10^4, 0.012]$



4. LSPE for Average-Reward (Part IV)

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Why LSPE for average-reward?

Discounted setting → Bellman error

$$\mathcal{B}^{\theta}(x, u) = \mathbb{E}[r + \gamma Q^{\theta}(x', \pi(x')) \mid x, u] - Q^{\theta}(x, u)$$

Average-reward setting → Poisson error

$$\mathcal{P}^{\theta}(x, u) = \mathbb{E}[r + Q^{\theta}(x', \pi(x')) \mid x, u] - \eta^{\theta} - Q^{\theta}(x, u)$$

Key difference

- No discount factor $\gamma \rightarrow$ need to subtract average reward η
- LSPE minimizes mean squared Poisson error
- Jointly estimates θ (Q-parameters) and η (average reward)

Least-Squares Policy Evaluation

Objective Find $\theta = [\theta_Q; \eta]$ that minimizes:

$$\min_{\theta} \mathbb{E} \left[\left(\mathcal{P}^{\theta}(x, u) \right)^2 \right]$$

LSPE solution Define $\varphi_k = [\mathbb{E}[\psi'] - \psi_k; -1]$, then solve:

$$A\theta = b \quad \text{where} \quad A = \mathbb{E}[\varphi\varphi^{\top}], \quad b = -\mathbb{E}[\varphi \cdot r]$$

Implementation

- Monte-Carlo approximation of $\mathbb{E}[\psi(x', \pi(x'))]$
- Regularization: $(A + \lambda I)$ for stability
- Quadratic basis ψ : 14 features (no constant term)

Q8.5–Q8.6: Evaluating

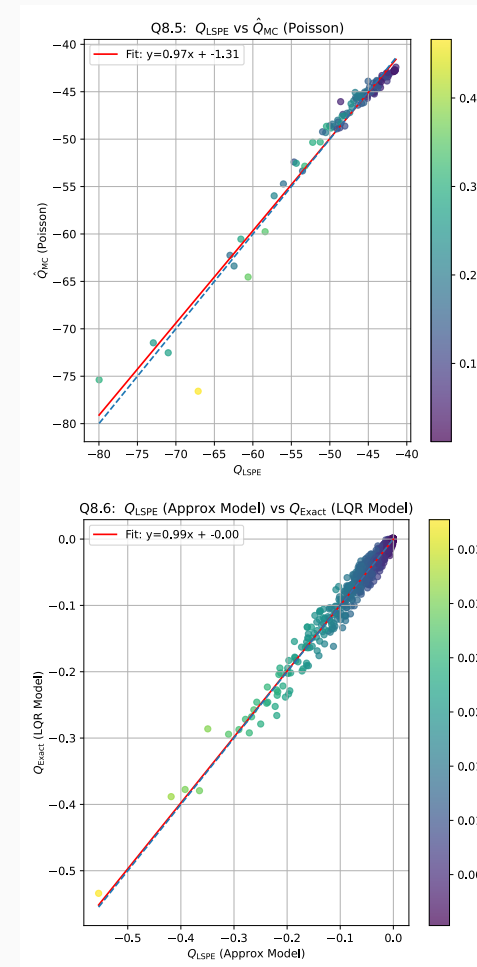
π_{cl}

Q8.5: True system

- LSPE matches Monte-Carlo $\hat{Q}$
- Validates the Poisson formulation

Q8.6: LQR approximate model

- LSPE matches exact Q_{LQR}
- Confirms correctness of implementation



4. LSPE for Average-Reward (Part IV)

— LSPE Results: Policy Evaluation

LSPE+PI algorithm

Iterate:

1. **Data collection:** Explore with current policy + noise ($\sigma_{\text{exp}} = 0.02$)
2. **Evaluation:** LSPE $\rightarrow \theta_Q, \eta$
3. **Improvement:** Greedy policy from Q^θ

Greedy policy extraction For quadratic $Q(x, u) = \dots + b(x)u + au^2$:

$$u^* = -\frac{b(x)}{2a}$$

Two settings tested

- Q8.7: Train & evaluate on **true system**
- Q8.8: Train on **LQR model**, evaluate on true system

4. LSPE for Average-Reward (Part IV) — 4.3 LSPE + Policy Iteration (Q8.7–Q8.8)

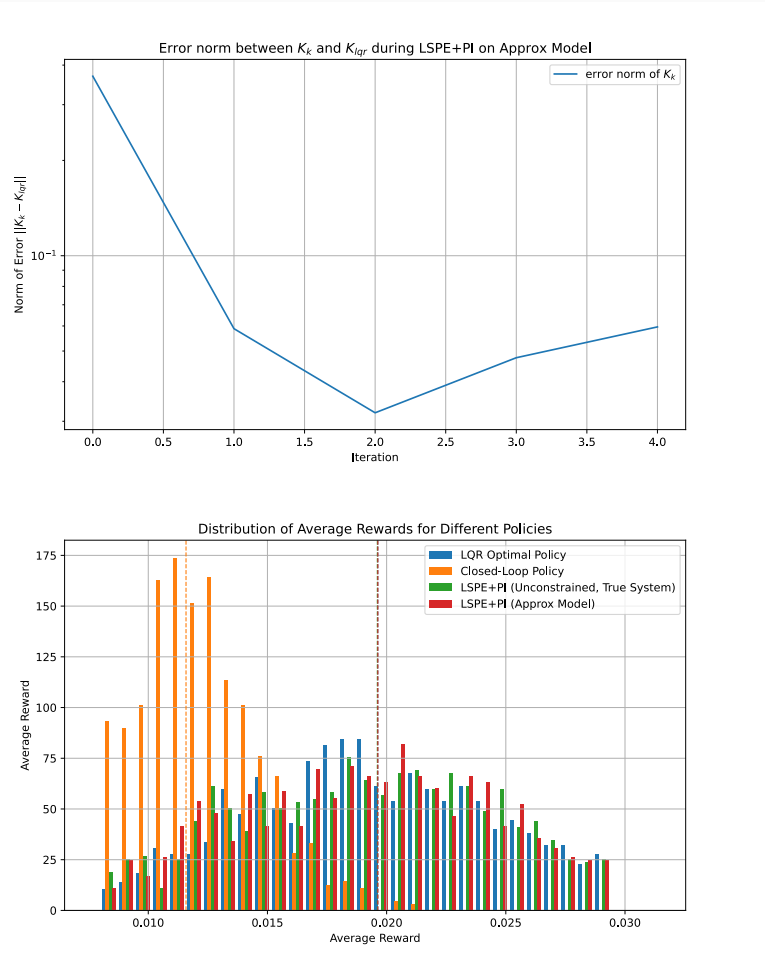
Convergence and performance

Q8.8: LQR model

- Gain K converges to K_{lqr} in *few* iterations
- Confirms LSPE+PI recovers optimal LQR policy

Q8.7: True system

- Similar performance as K_{lqr}



4. LSPE for Average-Reward (Part IV) — LSPE+PI Results (Q8.7–Q8.8)

5. Model Comparison

5. Model Comparison

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Policy	Model	const	Avg Reward	Avg Reward (unc)	Comments
π_{cl}	True	✗	0.005796	0.011451	
CMA-ES	True	✓	0.009654	—	
π_{lqr}	LQR	✗	0.009693	0.019467	$\rightarrow K_{lqr}$
MPC ($\mathcal{N} = 10$)	LQR det	✓	0.010497	—	higher compute
E-PIA	LQR	✗	0.009651	0.019584	$\rightarrow K_{lqr}$
π_{lspe}	True	✗	—	—	near-LQR on true model
Q_λ	True det	✓	0.000011	0.000019	
π_{lspepi} approx	LQR	✗	0.009833	0.019534	$\rightarrow K_{lqr}$
π_{lspepi} true	True	✗	0.009763	0.019452	$\rightarrow K_{lqr}$
π_{lspepi} true constr	True	✓	0.009725	—	

COMPUTED on True system :

- $T = 200$
- $N = 5000$ ($N = 100$ for MPC)
- $x_0 = (0 \ 0 \ 0)$

Comparison :

- mpc is better (but tested only at N=100)
- π_{lqr} , Q_λ on LQR model all methods converge to K_{lqr}
- method trained with quadratic psi tend to $\approx$ LQR performance :
 - models rewards small variance => near zero (near our LQR approximation with c_quad)

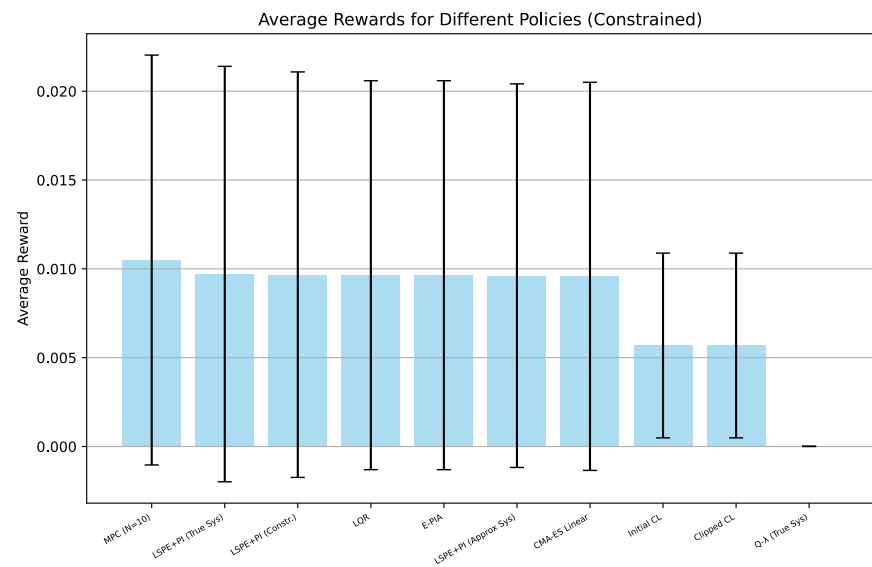


Figure 14: Average reward comparison

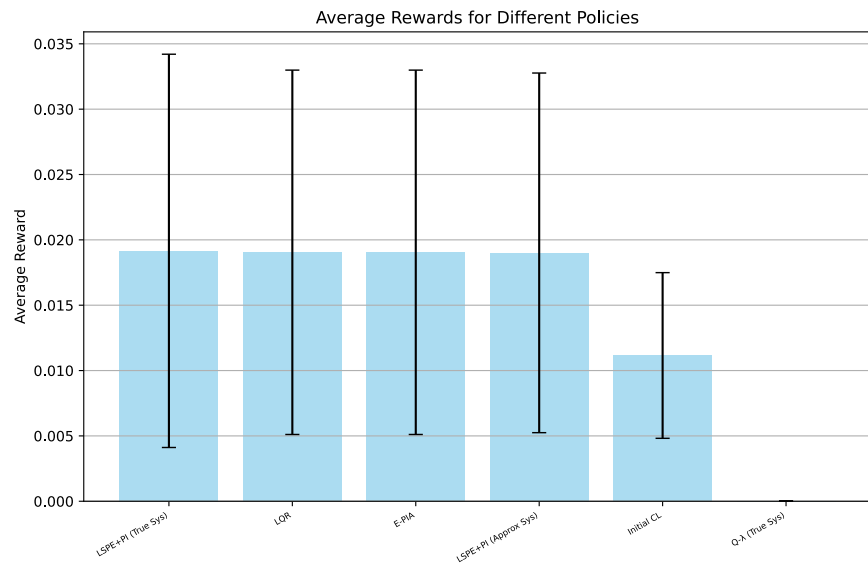


Figure 15: Average reward comparison (unconstrained)

MPC reward computed on $N = 200$ samples due to high compute cost

Design choices & difficulties

Design choices

- enforce feasibility via clipping; compare with MPC
- CMA-ES for noisy/non-smooth policy optimization
- quadratic features for LQR-consistent learning; higher-degree for true utility
- MPC horizon $\mathcal{N} = 10$ as compute/reward trade-off
- exploration: Gaussian around $K_{cl}x$ with tuned variance

Difficulties + how we addressed them

- saturation episodes reduce reward $\Rightarrow$ compare “fairly” (all constrained vs all unconstrained)
- Q- λ instability at large $\lambda \Rightarrow$ reduce learning rate α
- average-reward evaluation needs $\eta \Rightarrow$ Poisson/LSPE directly estimates it

Thank you for your attention !

Aymeric Couplet and Lucas Ahou

6. Appendix

6. Appendix

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Baseline distributions

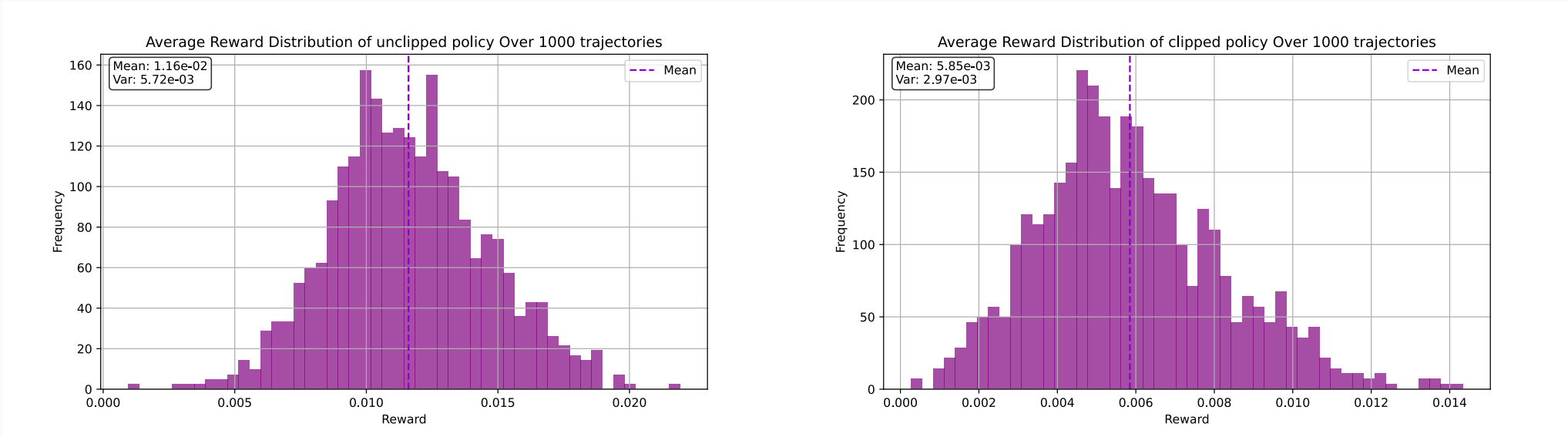


Figure 16: Unclipped baseline

Figure 17: Clipped baseline

LQR distributions

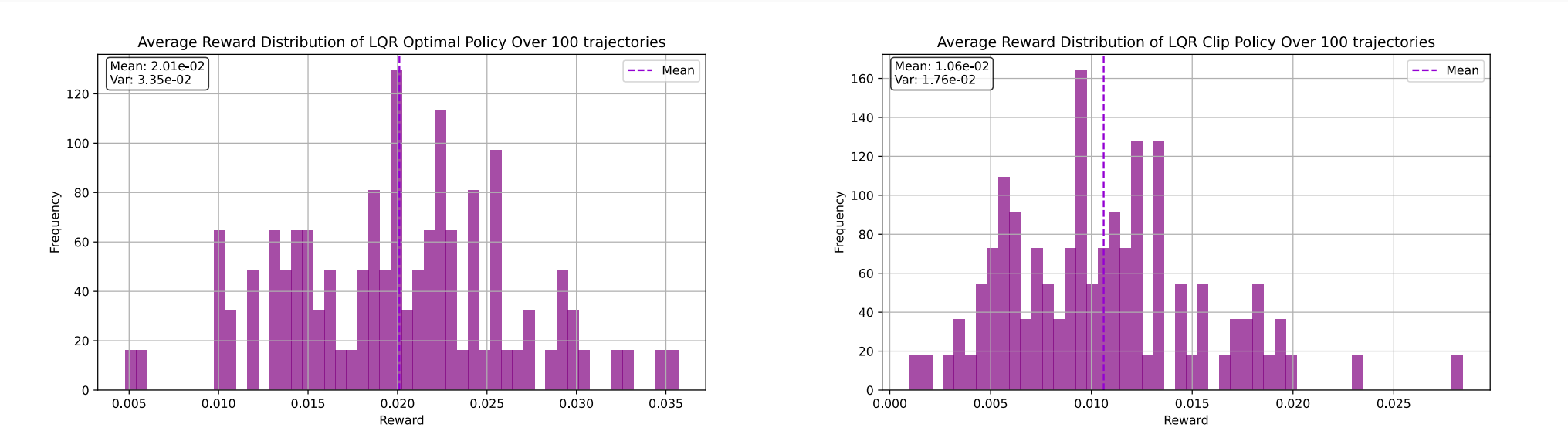


Figure 18: Unclipped LQR

Figure 19: Clipped LQR

MPC extra plots

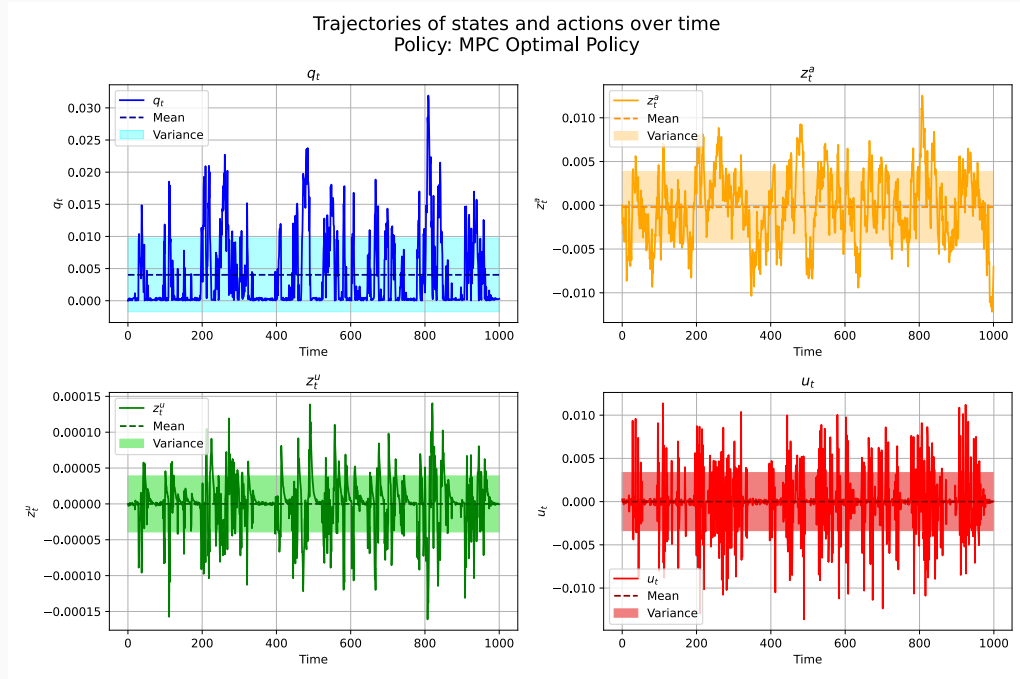


Figure 20: Example MPC states/actions

E-PIA convergence

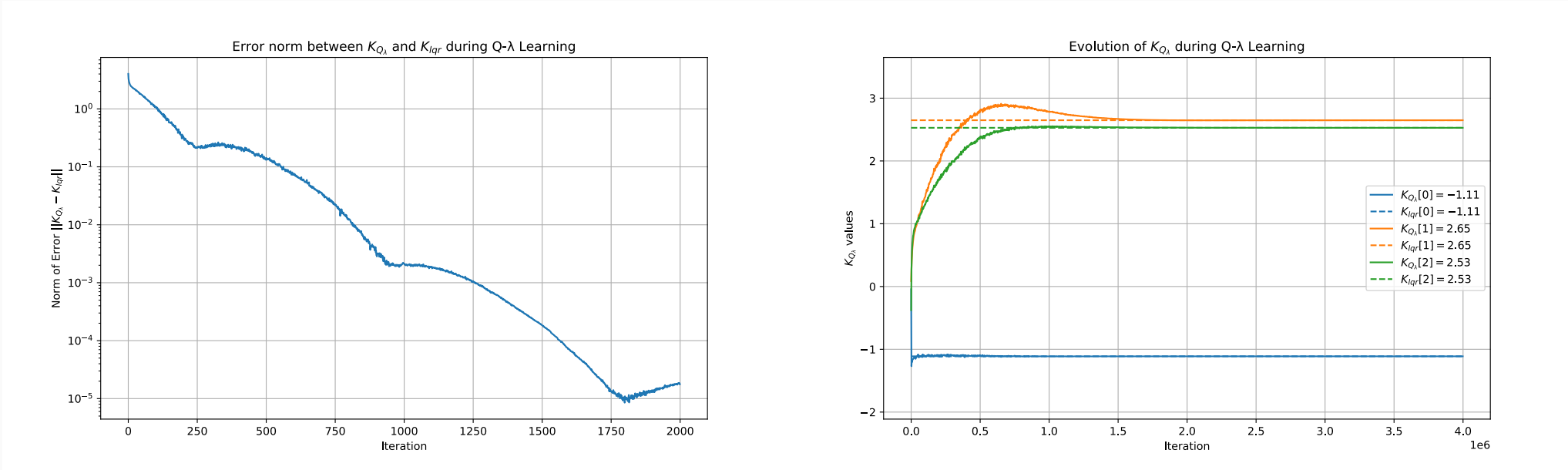


Figure 21: Error norm to K_{lqr}

Figure 22: K_k components

LSTD evaluation

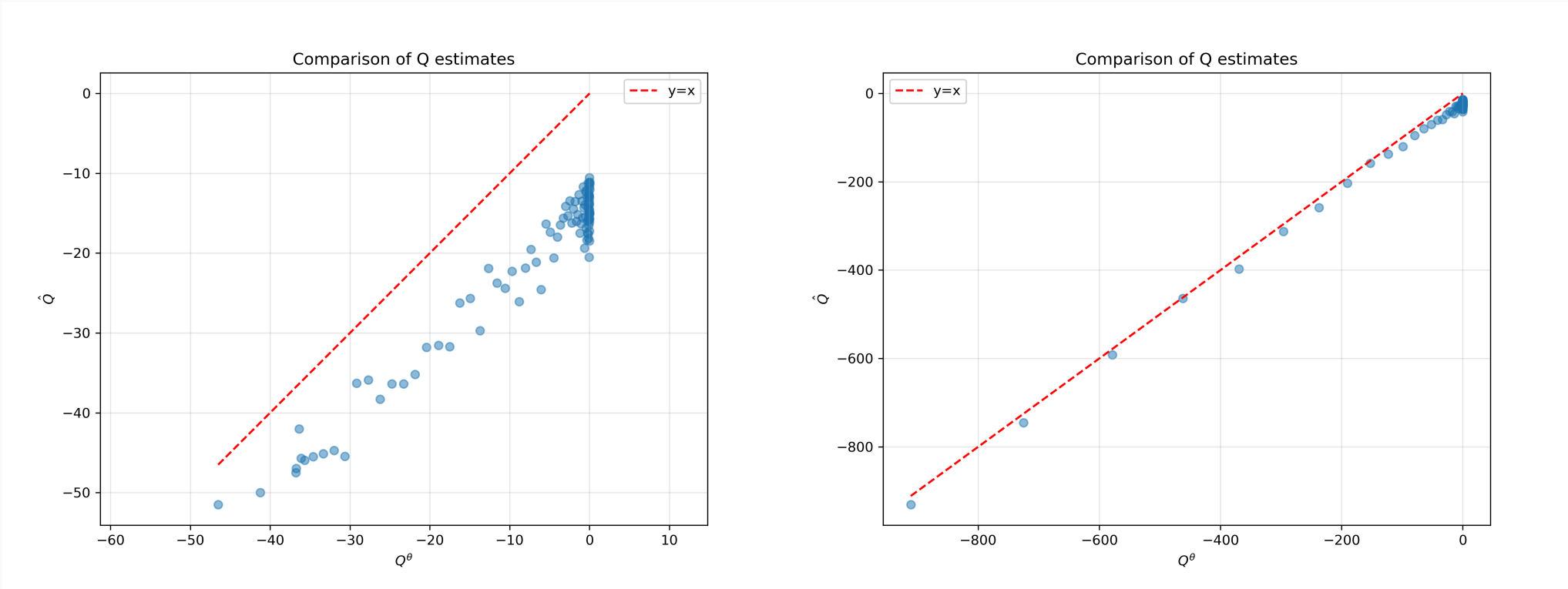


Figure 23: True system: Q^θ vs $\hat{Q}$

Figure 24: LQR model: Q^θ vs $\hat{Q}$

- 6. Appendix
 - Appendix: LSTD evaluation plots (Q6.3–Q6.4)

LSPI: convergence + constraints

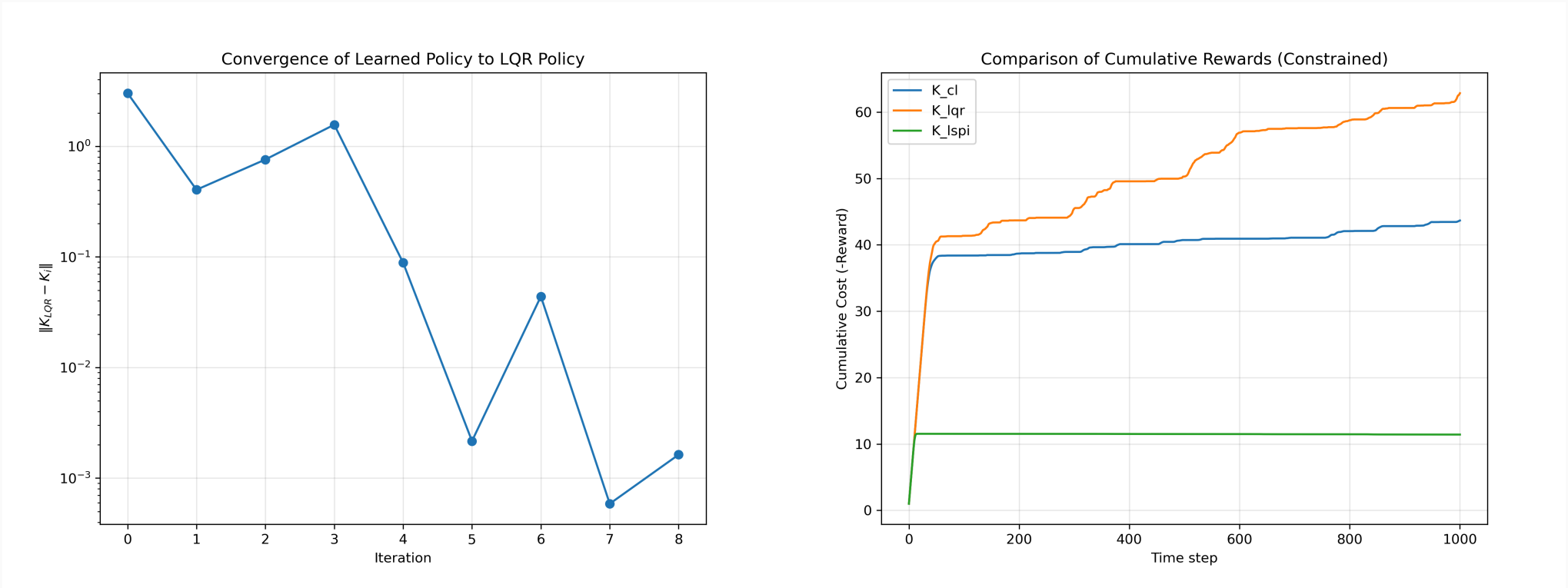


Figure 25: LSPI gain converges to K_{lqr}

Figure 26: Constrained PI can reduce reward

6. Appendix
— Appendix: LSPI convergence and constrained PI (Q6.6–Q6.7)

LSPE+PI with constrained improvement

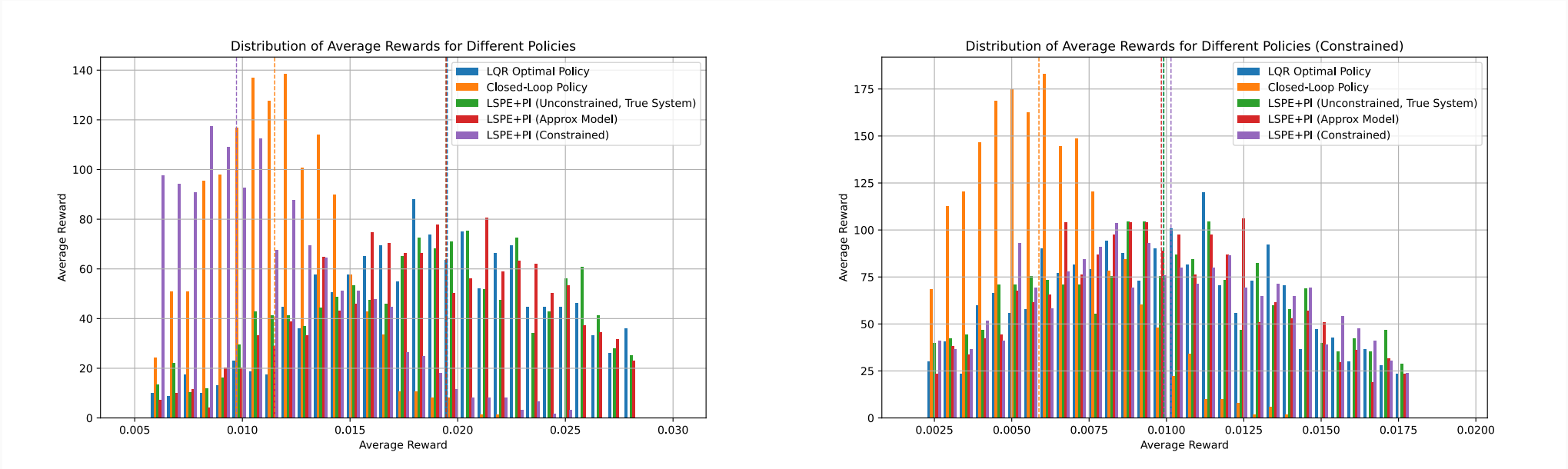


Figure 27: Unconstrained comparison

Figure 28: Fair comparison: all constrained

no good use of this

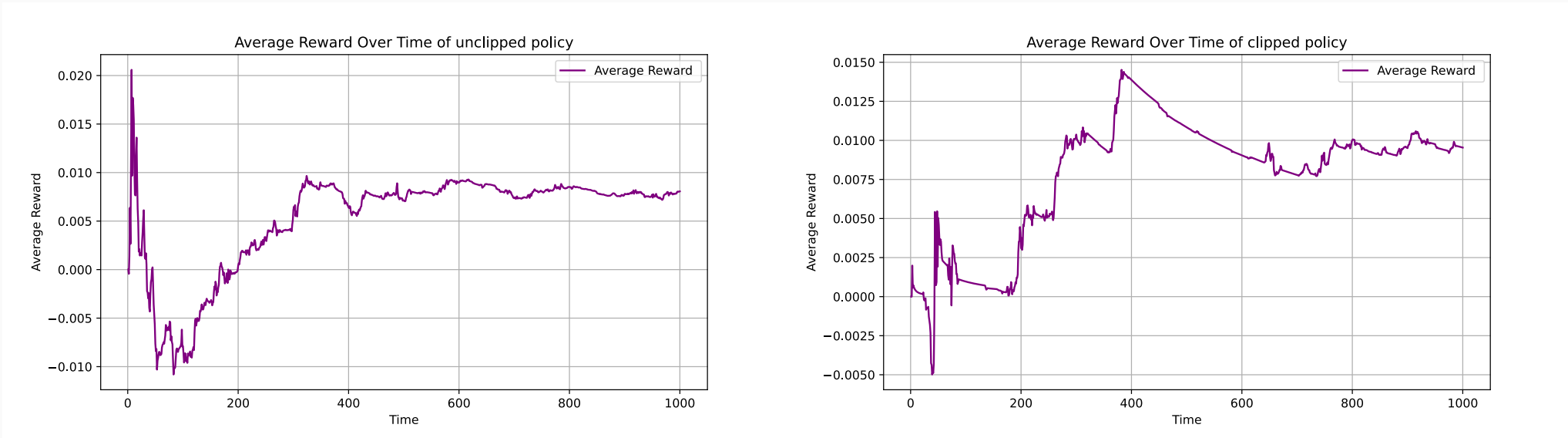


Figure 29: Rewards of π_{cl}

Figure 30: Rewards of π_{pcl}